

The AD Management Dashboard: Get a Grip

The creation of a “dashboard” process is imperative for AD organizations that wish to remain competitive and proactively manage application delivery.

Core Topics

Applications Development: Applications Development Process Management; Applications Development Project Management

Key Issues

What strategies, processes and techniques will assist AD organizations in reducing their exposure to project failures?

Which technologies and vendors will provide effective process management solutions in support of AD methodology delivery?

Applications development (AD) managers work within the context of a multiconstraint model. Constraints include (but are not limited to): budget, resource capacity, skillsets, customer expectation and technology infrastructure. Against the background noise of such common complaints as “I had to have it yesterday,” “do more with less” and “just one more change, please,” AD managers can gain efficiency, productivity and some peace of mind by setting up and monitoring an AD management “dashboard.”

A dashboard comprises a set of organization-specific metrics pertinent to AD and delivery, and lets AD managers to “manage by exception” (i.e., take action when a tolerance range has been exceeded). Data gathering processes must be set up with senior IS/AD management support, and focus initially on the AD group’s key areas of concern. A comparative benchmark can help identify areas of relative weakness, but typically, management can identify key goals such as improved application quality, reduced cancellations, better estimating and cutting maintenance costs.

Procedures for collecting specific data are then built into the current process, with the conviction that they are key to problem prevention and process improvement, and will yield substantial payback. These procedures should normally be executed by an authorized project manager, perhaps working out of a project office. Tools (e.g., estimating, testing or time collection) will support these procedures. Savings gained by assessing and applying the knowledge derived from the new procedures should be quantified and publicized.

In previous research, we have indicated productivity, cost and quality measurements based on function points (see *Research Note* KA-04-5051, and Note 1). However, other categories of

Note 1

Process Metrics for AD

Productivity (Development/Enhancement):
Function points per staff hour

Productivity (Support): Function points supported per full-time maintenance programmer

Quality (Product): Defects per function point found during the warranty period (usually 30 to 90 days after production)

Cost (Development/Enhancement): Cost per function point

Cost (Support): Cost per function point

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Note 2

Examples of Specific Metrics

Project Management:

- Milestone achievement
- Earned value
- Critical-path slippage
- Scope creep (i.e., changes)

Resource Management:

- Resource loading
- Billability
- Overtime

Reuse:

- Time for catalog searches/analysis
- No. of artifacts incorporated
- No. of new artifacts harvested

Quality:

- Defect arrival rate
- Defect removal rate
- Maintenance expense

metrics may be more immediately useful while AD groups develop function-point skills. Note 2 gives some examples of specific metrics in the following categories, but an organization may require more, or a slightly different set.

Project Management: Where do high-priority projects stand? Project delivery rates directly impact customer satisfaction, IT's value to the business, the competitive edge, market share and profitability.

Resource Management: Where are people, and what are they doing? Tracking skillsets, learning rates, availability and performance will enhance a manager's ability to optimize resources across efforts, consequently impacting application delivery, cost and quality.

Reuse: What is the leverage factor? How often is the wheel being reinvented? Reuse should involve all AD elements (e.g., methods, techniques, objects, components, source code, project plans and estimates). Reuse achievement rates can directly impact project delivery, time to market, quality (assuming a previously quality-assured element) and competitive advantage.

Quality: Is it being delivered? Quality can have multiple meanings based on constituency — for example, delivered on time within budget, a highly intuitive user interface, defects discovered during implementation period, application response times, the volume of change requests, or maintenance costs.

Note 3

Examples of "Feeder Tool" Categories

- Project planning and scheduling
- Resource management
- Time-sheet tracking
- Software configuration management
- Defect tracking
- Metadata repositories
- Test management

Note 4

Examples of Consolidator Tools

Centerline's Acqua

Platinum Technology's ADvisor

Micro-Frame's Portfolio

Artemis' Global View

PlanView's SnapShot

ABT's Results Management

Numerous tools (see Note 3) exist to help generate metrics data, and in many cases, are already in place within organizations. Additionally, a new market is emerging containing "consolidator-type" tools that gather these metrics, conduct further analysis and then present results to managers (see Note 4). In fact, some are, or will shortly be, leveraging Web-based push technology to generate real-time alerts. Although promising, the actual implementation of these tools should be secondary to the creation of the dashboard process itself, and identification of its key metrics.

Bottom Line: Only "reactive" decision making is possible, with limited visibility into project status, resource utilization, reuse activities and quality levels. Given the criticality of today's enterprise initiatives (e.g., year 2000, E-commerce, enterprise resource planning implementations and the euro) and a growing intolerance for failure, AD managers need a more proactive approach. The creation of a dashboard, validated against business requirements for AD, is an absolute imperative for AD organizations that wish to remain competitive and proactively manage application delivery.

